

CS2422 Assembly Language & System Programming

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Today's Topics

- Learning Assembly by Examples.
 - Program Formats
 - Some Simple Instructions
 - Assemble and Execute

Study Guide

- Textbook Sections 3.1 to 3.3.

Example: Adding 3 Integers

```
TITLE Add and Sub
; This is a comment
INCLUDE Irvine32.inc

.code
Main proc
    MOV EAX, 10000h
    ADD EAX, 40000h
    SUB EAX, 20000h
    CALL DumpRegs
    EXIT
Main endp
End Main
```

Format of Program

- Identifiers are case insensitive.
- Comments follow ';' character.
- Sections: Data, Code, and Stack.
- Hexadecimal Numbers have 'h' suffix.

- Directives vs. Instructions.
 - Use Directives to tell assembler what to do.
 - Use Instructions to tell CPU what to do.
- Procedure defined by:
 - [Name] PROC
 - [Name] ENDP
- Instruction Format:
 - LABEL (optional), Mnemonic, Operands

I/O

- Not as easy as you think, if you program it yourself.
- We will use the library provided by the author of the textbook.
- Two steps:
 - Include the library (Irvine32.inc) in your code.
 - Call the subroutines.

Assemble and Run!

- The required software comes with the book:
 - MASM: Microsoft Macro Assembler
 - CodeView: A Debugger
 - Irvine32.inc: I/O procedures

Figure 3–1 Assemble-Link-Execute Cycle.

